

Application of Distributed FIR filter based on FPGA in the analyzing of ECG signal

LI Nian-qiang
the University of Jinan 250022
JI Nan, China
lse_linq@ujn.edu.cn

Hou Si-Yu
the University of JI Nan 250022
JI Nan, China
heou_l@sina.com

Cui Shi-Yao
the University of JI Nan 250022
JI Nan, China
lse_cuisy@ujn.edu.cn

Abstract—The correct diagnosis of cardiovascular disease depends on the accuracy of ECG acquisition. This article describes the principle of distributed algorithms and applied to the ECG signal FIR filter design methods, combined with Altera's Cyclone series chip made to achieve FIR digital filter design. In the Quartus II and the Matlab environment to complete the simulation and synthesis, get good results.

Keyword—ECG signal; Distributed Algorithm; FIR filter; FPGA

I. INTRODUCTION

Currently, cardiovascular disease remains a threat to human health and safety of one of the major diseases. With the rapid economic development as well as changes in lifestyle, cardiovascular disease morbidity and mortality are showing a rising trend. The correct diagnosis of cardiovascular disease depends on the accuracy of ECG acquisition.

Collected from the body surface ECG signal is very weak, usually only 0.05 mV ~ 5mV. Spectrum scope mainly focus on 0.05 ~ 100Hz, Ecg signals through acquisition system by simulation signals into digital signals. ECG signal is processed, besides contain useful signals, and also contain many interference signals. Including the 50Hz power frequency interference, the frequency is about 0.1Hz ~ 0.3Hz baseline drift and interference signals above a frequency of 100Hz. Usually, ECG detected in diagnostic mode only requires the signal frequency below 150Hz. So with pure ECG signals, we should design a low-pass filter.

Finite Impulse Response (FIR) filters are the most basic digital signal processing system components. It is at any rate the frequency with a strictly linear phase frequency. There is no input to output feedback, which is a stable system.

Compared with the traditional algorithms, distributed algorithms can greatly reduce the size of the hardware circuitry to improve the circuit speed of execution. Recently, as is widely used in recent FPGA digital signal processing, distributed algorithm in the FPGA to achieve the FIR digital filter to become very real needs.

II. FIR FILTER ALGORITHM OF DISTRIBUTED

A. FIR filter theory

The basic structure of FIR filter is like a sub-section of the delay lines. Each section of the output of the weighted

cumulative, you can get the output of filter. Mathematical expression is:

$$y(n) = \sum_{k=0}^{L-1} h(k)x(n-k) \quad (1)$$

The corresponding pulse sequence is a unit of output $h(k)$ and input $x(n)$ of the convolution. Through the convolution relationship diagram can be directly drawn structure. It is called direct-type structure. As shown in Fig.1. Equation (1), $y(n)$ is the n -time filter output, L is the number of FIR filter taps, $h(k)$ is the k -class tap coefficient (unit impulse response), $x(n-k)$ for k -tap delay the input signal. Figure 1 shows the N -order direct form FIR filter structure. It is committed by a "tapped delay line" adder and multiplier set constitutes. Each $y(n)$ the output should be carried out N times multiplications and $N-1$ times to calculate the product and the addition.

FIR digital filter transfer function as follows:

$$H(z) = \sum_{k=0}^{N-1} h(k)z^{-k} \quad (2)$$

In which, $h(k)$ is the filter coefficients. We can see from the diagram, FIR filter coefficients determined. By the use of MATLAB can be easily derived FIR filter coefficients.

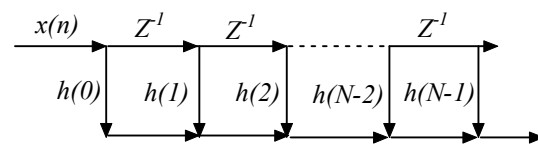


Figure 1. Convolution FIR filter structure

B. The theory of distributed FIR filter

Distributed arithmetic was first proposed by Croisier in 1973. But until Xilinx invented FPGA lookup tables structure, distributed arithmetic was widely applied in the calculation of the product.

Filter equation is rewritten as follows:

$$y(n) = \sum_{k=0}^{L-1} h_k x_k(n) \quad (3)$$

Input data $x_k(n)$ can be used to represent $B + 1$ bit complement, that is:

$$x_k(n) = -2^B x_{k,B}(n) + \sum_{b=0}^{B-1} x_{k,b}(n) 2^b, x_{k,b}(n) \in [0, 1] \quad (4)$$

In equation (4), B for data bits wide, $x_{k,b}(n)$ on behalf of $x_k(n)$ the b -bit, $x_{k,B}(n)$ is the sign bit. Be (3) into (2) were:

$$y(n) = \sum_{k=0}^{L-1} h_k \left[-2^B x_{k,B}(n) + \sum_{b=0}^{B-1} x_{k,b}(n) 2^b \right] \\ -2^B \sum_{k=0}^{L-1} h_k x_{k,B}(n) + \sum_{b=0}^{B-1} 2^b \cdot \left[\sum_{k=0}^{L-1} h_k x_{k,b}(n) \right] \quad (5)$$

By the equation (5) can be seen that entry in square brackets mean that all delayed the first b -bit input data and filter coefficients of each tap $h_0, h_1, h_2, \dots, h_{L-1}$, for the corresponding multiplication operator, and summation. Therefore, you can build a look-up table through the internal FPGA. In advance of all possible "and value" stored in the table, The actual calculation of the process of using all the input data into the corresponding bit combination of the vector as the address of the look-up table to address, it can direct access to this and value, the index part of the sum of the results of the corresponding bit of the right, through the shift to achieve, do not take up logic resources.

C. Distributed FIR filter implementation structure

From the above formula in order to achieve distributed algorithm is a multiply-accumulate operation for the purpose of computing methods. The traditional algorithm is different from multiply-accumulate operations. The Difference is that the implementation of the order of operations in different parts of the plot. Distributed algorithm in the completion of multiply-accumulate function is to input data by the corresponding bits of each partial product generated by the sum of advance part of the formation of the corresponding product, and then add to the accumulation of various parts of the formation of the final result. The traditional method is to have all of the product has generated the sum after the completion of multiply-accumulate operations.

Using a distributed algorithm, if the filter coefficients too much, then the LUT size will be very large, with a single look-up table is unable to perform full-word operations, and to find a way to find the partition table. We can consider two aspects of look-up table partitioning: First, reduce the size of lookup table. Second, improve the DA algorithm speed.

In order to reduce the size of lookup table, then large look-up table can be used to break down for small look-up table method to reduce the logic resource consumption. Since FIR filters are linear filters, so low-order filter output can be added together, and thus the definition of a high-order filter output. For example, 16 inputs LUT can be divided into four parallel input of the LUT. If you add lines, change in this structure does not reduce the speed, it can greatly reduce the design size. Dashed part of the said pipeline register. As shown in Fig. 2.

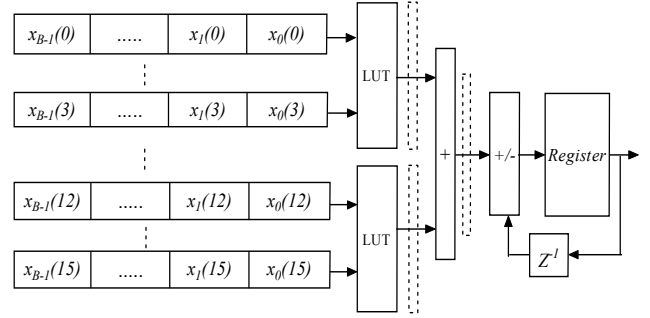


Figure 2. Simplify the structure of the scale of the distributed algorithm

In addition, you can use another way to improve computing speed. That is, the input field is divided into K sub-field and then in parallel to deal with these sub-fields. Computing speed will be split with the table number and turned many times, but this method requires K times the look-up table, resulting in storage demand and costs skyrocketed. In practice, you can combine these two approaches, taking into account the system's resources and performance, in order to achieve optimum.

Compared with the traditional algorithms, distributed algorithms using FPGA digital signal processing has played an important role, it greatly reduces the size of the hardware circuit, it is easy to implement pipelining technology to improve the circuit speed of implementation, greatly improved signal processing efficiency.

If you construct a LUT, the table stores all the possible combinations of brackets values, you can through all the input variables corresponding to the combination of vector-bit addressing of the table. Construction of the table rules, such as shown in Table 1.

TABLE I. CONSTRUCTION RULES

$x_b(N-1)$	$x_b(N-2)$	...	$x_b(b)$	$x_b(0)$	Set the value of the LUT
0	0	...	0	0	$h(0) \cdot 0 + h(1) \cdot 0 + \dots + h(n-1) \cdot 0 = 0$
0	0	...	0	1	$h(0) \cdot 1 + h(1) \cdot 0 + \dots + h(n-1) \cdot 0 = h(0)$
b_{N-1}	b_{N-2}	...	b_l	b_0	$h(0)b_0 + h(1)b_1 + \dots + h(n-1)b_{N-1}$
1	1	1	1	1	$h(0) + h(1) + \dots + h(n-1)$

III. DESIGN AND IMPLEMENTATION OF FIR FILTER

A. Design specifications

Matlab is an interactive matrix-based software is widely used in teaching, research and engineering design and other fields. With the constant improvement in Matlab . Especially the Matlab signal processing toolbox launch, and now Matlab has become a digital signal processing applications in analysis and design of the main tools for simulation.

In order to cut-off frequency of ECG waveform control 110Hz, using MATLAB to design a sampling frequency of 48kHz , the filter cut-off frequency of 150Hz of the FIR filter.

Usually, linear phase FIR filter using window function method. MATLAB window function used here to design. The basic idea of the design window function is to select a ideal frequency selective filters. And then truncate the impulse response to obtain a linear phase and causal FIR filter. According to the given filter specifications, Selected Kaiser window design, 16-order FIR digital low-pass filter characteristic parameters are as follows:

$$\begin{aligned} h[0] &= h[15] = 0.028219 \\ h[1] &= h[14] = -0.0073159 \\ h[2] &= h[13] = -0.049899 \\ h[3] &= h[12] = -0.061658 \\ h[4] &= h[11] = -0.01404 \\ h[5] &= h[10] = 0.089426 \\ h[6] &= h[9] = 0.20909 \\ h[7] &= h[8] = 0.28894 \end{aligned}$$

B. Distributed FIR filter of FPGA implementation

FPGA-based design of FIR filters, mainly by the multipliers, adders and shift registers to achieve. Its core is a multiply-accumulate function. Look-up table approach is to use lookup tables instead of multipliers, thereby to increase the computing speed and reduce resource consumption.

The use of programmable logic device to achieve FIR filters, due to hardware implementation of the parallel structure, particularly suitable for demanding real-time occasions. This article uses Alter company Cyclone III chips to design the FIR filter.

As calculated from the MATLAB, the coefficients $h(n)$ is a set of floating-point value, and the FPGA device is only undertake fixed-point value calculation, so to conduct floating-point to fixed-point conversion. With 8-bit binary complement representation $h(n)$ values are as follows:

$$\begin{aligned} h[0] &= h[15] = 00001110 \\ h[1] &= h[14] = 10000100 \\ h[2] &= h[13] = 10011010 \\ h[3] &= h[12] = 10100000 \\ h[4] &= h[11] = 10000111 \end{aligned}$$

$$h[5] = h[10] = 00101110$$

$$h[6] = h[9] = 01101011$$

$$h[7] = h[8] = 10010100$$

Distributed Arithmetic FIR filter block diagram shown in Fig.3. How It Works: *CTRL_ALL* for each module provides clock and reset, and the necessary control signals. The data *DAT_IN* input shift Register with *CLK1_16*, data in the shift ram in the shift clock is *CLK*, *TAP* interval of 16, i.e the first one *CLK1_16* in *TAP0* have data. The 16-CLK choose to keep the data in 16 different bits in the *CLK* in MUX16x1, and then into the lookup table. In the second *CLK1_16* in, *TAP0* and *TAP1* have data output. The '*b*' is the left-digit, $0 \leq b \leq B$, '*B*' is the input data bit-width, $B=16$. When $b = 16$, the addition and subtraction device.

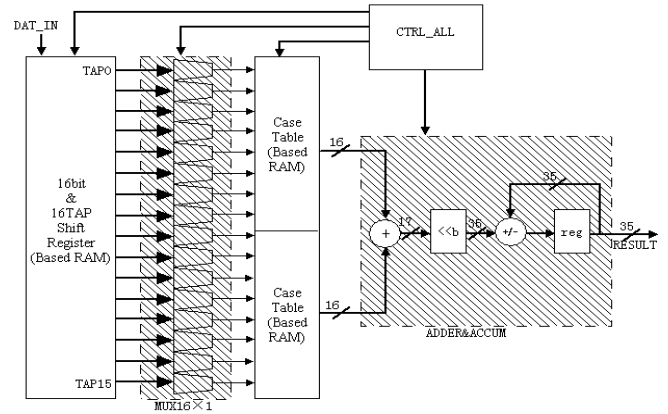


Figure 3. Distributed FIR Filter Block Diagram

Distributed FIR filter according to the structure of distributed algorithms, using Verilog HDL language, Quartus II 8.0 platform to be simulated. As the input data to pre-add, it must consider the overflow. Generally use the sign extension method to prevent overflow, that is, the input data is a sign-bit extensions, so that the data expanded more than the original 1-bit input data. Using modelsim simulation waveforms shown in Fig.4.

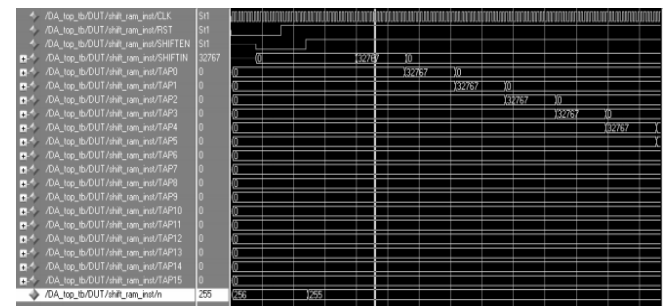


Figure 4. Waveform Simulation diagram

IV. FILTERING EFFECT

The use of the filter designed in this paper for some serious noise on the ECG signal filtering, comparison of the simulation shown in Fig.5 and Fig.6. In Fig.5 diagram is the

ECG signal with a high-frequency interference, Fig.6 plan is the result obtained by the filter designed in this paper, Read from the filtering effect, because the FIR filter designed in this paper uses a distributed algorithm, on the ECG in the high-frequency interference have a stronger filtering ability, better filtering effect and get a more smooth and more realistic ECG curve.

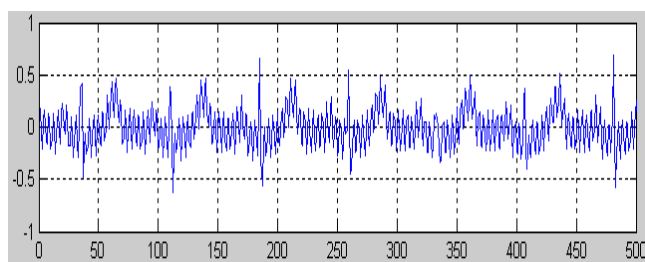


Figure 5. The ECG signal with noise

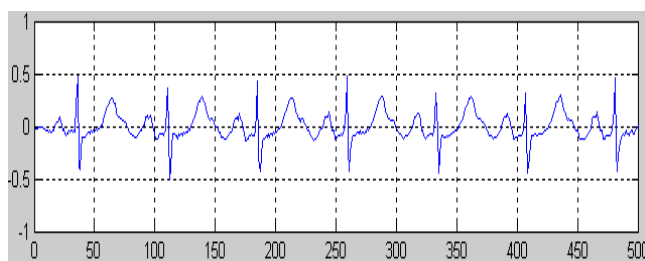


Figure 6. ECG signal after filter

V. CONCLUSION

This paper mainly studies the distributed algorithm for FIR filter. ROM look-up table method based on a distributed algorithm uses a wealth of RAM within the FPGA module. The advantages are hardware resources occupied by a small, compact structure and high operating frequency. This article also used Verilog HDL hardware description language in the FPGA to design a distributed FIR filter. And use Quartus II 8.0 to conduct functional simulation and timing simulation. By changing the filter order and the lookup table in the coefficient of this design, it can be applied to achieve high-pass, low pass and band stop filter, and better portability. And applied to the ECG signal pre-processing, the realization of the ECG signal filtering high-frequency interference. It works well to maintain the original signal and has important practical value.

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